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Review Article

Meta-analysis of the association between prenatal antibiotic exposure and risk of childhood attention-deficit/hyperactivity disorder

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ABSTRACT

Background: Studies examining the association between maternal antibiotic use during pregnancy and the risk of childhood attention-deficit/hyperactivity disorder (ADHD) have yielded inconsistent results. This meta-analysis synthesizes available evidence to establish a more comprehensive understanding of this association.
Methods: Observational studies published through October 2, 2024, were systematically searched from Cochrane Library, EMBASE, Web of Science, and PubMed databases. The Newcastle-Ottawa Scale was used to assess the quality of the included studies, and statistical analyses were performed using Stata version 15.0.
Results: Nine studies encompassing 6,180,434 participants were included. Prenatal antibiotic exposure was associated with an increased risk of childhood ADHD (HR = 1.15; 95 % CI = 1.09–1.22, $P < 0.0001$; OR = 1.28; 95 % CI = 1.21–1.35, $P < 0.001$). Both mid-pregnancy (HR = 1.11; 95 % CI = 1.02–1.22, $P = 0.282$) and late-pregnancy (HR = 1.07; 95 % CI = 1.02–1.12, $P = 0.521$) antibiotic exposure was associated with an increased risk of ADHD. Repeated antibiotic exposures was associated with an increased risk further (twice: HR = 1.13; 95 % CI = 1.10–1.16, $P = 0.856$; ≥3 times: HR = 1.21; 95 % CI = 1.17–1.24, $P = 0.166$).
Conclusion: This analysis identified a link between maternal antibiotic use during pregnancy and an increased risk of childhood ADHD. Large-scale, multicenter, well-designed prospective studies are needed to confirm these findings and adjust for potential confounders in evaluating the relationship between prenatal antibiotic exposure and childhood ADHD.

1. Introduction

Attention-deficit hyperactivity disorder (ADHD) is the most prevalent neurodevelopmental disorder and frequently coexists with other neurodevelopmental conditions, including autism spectrum disorder, intellectual disability, and tic disorders (Thapar and Cooper, 2016). This disease is characterized by inattention, impulsivity, and hyperactivity, and its prevalence continues to rise, imposing substantial burdens on individuals, families, and society (Posner et al., 2020). ADHD has multiple etiological factors. Although genetic factors are crucial in ADHD etiology, mounting evidence indicates the significance of environmental factors (Thapar et al., 2013). Among these, prenatal risk factors are significant contributors. Despite extensive research, its risk factors remain inconclusive (Kian et al., 2022). Notably, studies examining prenatal antibiotic exposure remain limited and yield conflicting results regarding the risk of ADHD in offspring, partly due to the inherent uncontrollability of exposure conditions.

Antibiotics are the primary therapeutic agents for the prevention and treatment of infections. While most antibiotics demonstrate minimal adverse effects during administration, they alter gut microbiota composition, which may result in short- and long-term health consequences (Ramirez et al., 2020). Antibiotic use during pregnancy remains common, accounting for 80 % of prescription medications in Western countries. Beyond disrupting maternal microbiota (Kuperman and Koren, 2016), prenatal antibiotic exposure also modifies the composition of gut microbiota in infants (Grech et al., 2021).

Substantial evidence demonstrates a complex interplay between gut microbiota and the nervous system (He et al., 2024). Alterations in gut microbiota can affect behavior through gut-brain axis mechanisms, potentially contributing to neurodevelopmental disorders (Wang et al., 2023) including ADHD. Studies have identified distinct gut microbial

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Abbreviations

ADHD	attention-deficit/hyperactivity disorder
PROSPERO	International Prospective Register of Systematic Reviews
MeSH	Medical Subject Headings
NOS	The Newcastle-Ottawa Scale
OR	odds ratios
RR	risk ratios
HR	hazard ratios
CI	confidence intervals
GBA	gut-brain axis
SCFAs	Short Chain Fatty Acids

compositions in ADHD patients compared with healthy controls (Liang-Jen et al., 2019; Jiang et al., 2018). Furthermore, gut microbiota from patients with ADHD affects brain function in mice (Tengeler et al., 2020), suggesting that microbial alterations may drive structural and functional brain changes. However, investigations into the relationship between prenatal antibiotic use and offspring neurodevelopmental disorders have yielded inconsistent results.

Given the increasing use of antibiotics, this meta-analysis aimed to synthesize available evidence to examine the link between maternal antibiotic use during pregnancy and the risk of childhood ADHD, providing evidence-based guidance for clinical practice.

2. Methods

This meta-analysis was prospectively registered in PROSPERO (International Prospective Register of Systematic Reviews) under registration number CRD42024539313.

2.1. Search strategy

The systematic review followed PRISMA and Cochrane guidelines. A comprehensive database search was performed across Cochrane Library, EMBASE, Web of Science, and PubMed through October 2, 2024. Medical Subject Headings (MeSH) terms and free-text keywords were combined to design search terms, including pregnancy, antibiotics, and childhood ADHD. Reference lists of relevant articles were manually screened for completeness. The search strategy is illustrated in Supplementary Materials.

2.2. Eligibility criteria

Eligibility criteria were established following the PECOS framework: 1. Population (P): Pregnant women; 2. Exposure (E): Antibiotic use; 3. Comparison (C): Pregnant women without antibiotic exposure; 4. Outcome (O): Childhood ADHD; 5. Study design (S): Cohort studies.

Exclusion criteria included: (1) animal studies or abstracts with incompatible study types, (2) non-pregnant populations, (3) pregnant women without antibiotic exposure, (4) no relevant outcome measures, and (5) non-English literature.

2.3. Literature screening

Two investigators independently examined the retrieved literature against predetermined criteria. Reports were imported to EndNote 20 to remove duplicates, and subsequently, titles and abstracts were checked to exclude irrelevant studies. The full texts of the remaining studies were downloaded and screened to determine eligibility. Disagreements during the literature screening process, if any, were addressed by a third investigator.

2.4. Data extraction

Two investigators independently extracted data from eligible studies, including first author, publication year, country, sample size, follow-up duration, gender ratio of children with ADHD, and age at diagnosis. Extracted data were cross-verified, with disagreements resolved through discussion with a third investigator.

2.5. Quality assessment

Two investigators independently assessed the study quality. The Newcastle-Ottawa Scale (NOS) was leveraged to appraise the quality of non-randomized studies based on the selection of study groups, comparability between groups, and outcome assessment. Studies could achieve a maximum score of 9 points. Those scoring $\geq$8 points were considered high quality, those with 6–7 points were at a moderate risk of bias, and those scoring $\leq$5 points were considered at a high risk of bias. Disagreements were addressed through team discussion.

2.6. Data synthesis

Statistical analyses were performed using Stata version 15.0. Effect measures included odds ratios (OR), risk ratios (RR), or hazard ratios (HR), with 95 % confidence intervals (CI). Results were presented using forest plots. Heterogeneity was evaluated via Cochran's Q test and I^2 values. Fixed-effects models were applied in the absence of significant heterogeneity; random-effects models were employed otherwise. Sensitivity analyses were executed to evaluate the stability of the results.

Publication bias was assessed through funnel plots and Egger's test. For significant publication bias, trim-and-fill methods were employed to evaluate its impact on the results. Statistical significance was defined as $P < 0.05$.

3. Results

3.1. Literature search

From databases, 781 articles were retrieved. After removing 58 duplicates, 723 articles remained. Following exclusion based on animal studies, incompatible populations or outcomes, 15 potentially eligible studies were identified. Six studies were eliminated due to inaccessible full texts ($n = 3$) or infeasible data extraction (n = 3). Finally, nine studies were ultimately included (Fig. 1).

3.2. Study characteristics

This analysis included nine studies (Lavebratt et al., 2019; Hamad et al., 2020; Fan et al., 2020; Aversa et al., 2021; Straughen et al., 2023; Njotto et al., 2023; Lin et al., 2023; Holmgaard et al., 2024; Jiyeon et al., 2024) encompassing 6,180,434 participants. Seven studies reported HR or RR (Lavebratt et al., 2019; Hamad et al., 2020; Fan et al., 2020; Aversa et al., 2021; Straughen et al., 2023; Lin et al., 2023; Jiyeon et al., 2024), while two reported OR (Njotto et al., 2023; Holmgaard et al., 2024). Two studies were conducted in Asian populations (Lin et al., 2023; Jiyeon et al., 2024), and seven in non-Asian populations (Lavebratt et al., 2019; Hamad et al., 2020; Fan et al., 2020; Aversa et al., 2021; Straughen et al., 2023; Njotto et al., 2023; Holmgaard et al., 2024). Publication dates ranged from 2019 to 2024. The baseline characteristics of the included studies are provided in Table 1.

Quality scores of included studies ranged from 6 to 9 points, indicating moderate to high methodological quality.

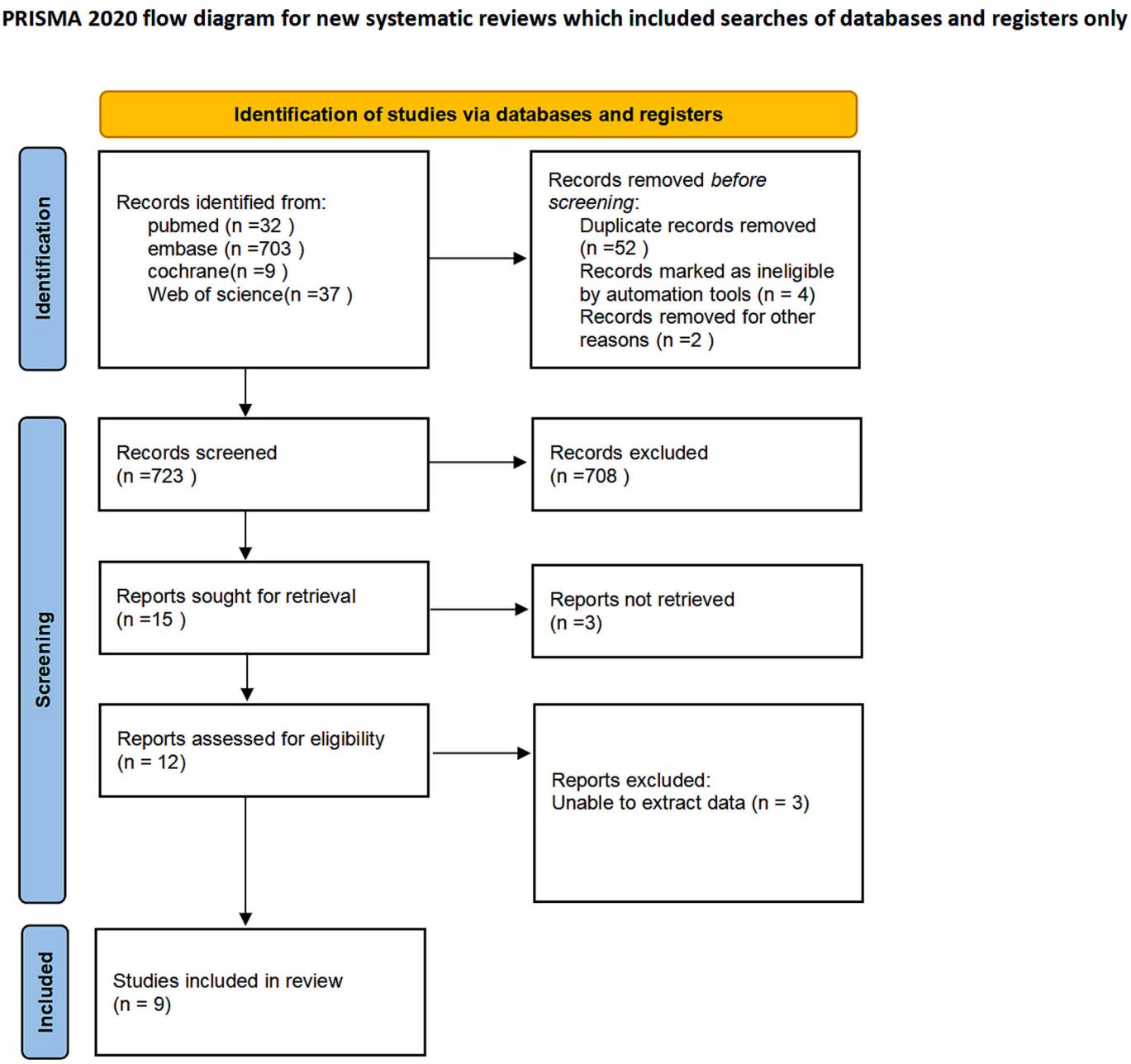


Fig. 1. Flow chart of the literature research.

3.3. HR analysis

3.3.1. Association between prenatal antibiotic exposure and childhood ADHD risk

Seven cohort studies providing HR or RR data (Lavebratt et al., 2019; Hamad et al., 2020; Fan et al., 2020; Aversa et al., 2021; Straughen et al., 2023; Lin et al., 2023; Jiyeon et al., 2024) were meta-analyzed using a random-effects approach due to substantial heterogeneity ($I^2 = 90.5$ %). Results revealed prenatal antibiotic exposure was associated with increased risk of ADHD in offspring (HR = 1.15; 95 % CI = 1.09–1.22, $P < 0.001$, Fig. 2). Subgroup analysis by geographic region showed no heterogeneity ($I^2 = 0$) in Asian populations (Fig. 3), suggesting geographic location as a potential source of heterogeneity.

3.3.2. Association between ADHD risk and antibiotic exposure during different gestational periods

Five studies examined specific gestational periods of antibiotic exposure. One study (Jiyeon et al., 2024) was excluded due to confounding psychiatric conditions in its outcomes. The meta-analysis was performed using fixed-effects models ($I^2 < 50$ % for all periods). The results showed that antibiotic exposure during mid-pregnancy (HR = 1.11; 95 % CI = 1.02–1.22, $P = 0.282$) and late pregnancy (HR = 1.07; 95 % CI = 1.02–1.12, $P = 0.521$) was associated with an increased risk of ADHD. Early pregnancy exposure (HR = 1.03; 95 % CI = 0.94–1.13, $P = 0.791$) was not associated with ADHD (Fig. 4).

3.3.3. Association between frequency of prenatal antibiotic exposure and childhood ADHD risk

Three studies (Hamad et al., 2020; Straughen et al., 2023; Lin et al., 2023) assessed how prenatal antibiotic exposure frequency influenced the risk of childhood ADHD. For single exposure, due to substantial heterogeneity ($I^2 = 55.4$ %), a random-effects model was used (Fig. 5). For two or more exposures, heterogeneity was low ($I^2 < 50$ %), and fixed-effects models were applied (Figs. 6, 7). The analysis showed that

Table 1
Basic characteristics and quality evaluation of included studies.

Author, year	Country	Sample size	Exposure assessment	Age at ADHD assess	NOS
Lavebratt et al. (2019)	Finland	990,098	The Drugs And Pregnancy database	2–18y	9
Hamad et al. (2020)	Canada	187,605	Antibiotic prescriptions	4–18y	9
Fan et al. (2020)	UK	726,274	CPRD	14y	9
Aversa et al. (2021)	USA	5632	The Rochester Epidemiology Project	nr	8
Straughen et al. (2023)	USA	555	Medical records	10y	7
Njotto et al. (2023)	Swedish	1,095,645	Prescribed Drug Registry	<11y	9
Lin et al. (2023)	Taiwan	906,942	DO and OO files	6y	9
Holmgaard et al. (2024)	Denmark	11,419	Maternal antibiotic prescriptions	nr	6
Jiyeon et al. (2024)	SouthKorea	3,163,206	The National Health Insurance Service data	nr	9

CPRD: the Clinical Practice Research Datalink; DO: inpatient order; OO: outpatient order; ADHD: attention-deficit/hyperactivity deficit.

repeated antibiotic exposures were associated with increased risk of ADHD: two exposures (HR = 1.13; 95 % CI = 1.10–1.16, P = 0.856) (Fig. 6) and three or more exposures (HR = 1.21; 95 % CI = 1.17–1.24, P = 0.166) (Fig. 7). Single exposure showed no statistically significant association (HR = 1.04; 95 % CI = 0.97–1.11, P = 0.106) (Fig. 5).

3.4. OR analysis

Meta-analysis of two studies reporting OR (Njotto et al., 2023; Holmgaard et al., 2024) showed no heterogeneity (I^2 = 0). The fixed-effects model was used. The results revealed that prenatal antibiotic exposure was associated with increased risk of childhood ADHD (OR = 1.28; 95 % CI = 1.21–1.35, P < 0.001) (Fig. 8).

3.5. Sensitivity analysis

Sensitivity analysis by sequentially excluding individual studies unraveled that the pooled HR estimates did not substantially change (Fig. 9), confirming the statistical robustness of the results. Additionally, Fan et al. (Fan et al., 2020) included infection among the adjusted confounders while other studies did not. After excluding this study and conducting sensitivity analysis, the results showed an HR of 1.16 (95 % CI = 1.09–1.23) (Fig. 11), with no significant difference compared to the original analysis. This finding suggests that inclusion of infection among the adjusted confounders may not substantially affect our results. However, given the limited data available for this analysis, these findings should be interpreted with caution and require further confirmation.

3.6. Publication bias

Funnel plots were used to assess potential bias for the seven studies reporting HR Funnel plots were asymmetrical (Fig. 10). Egger's test indicated no significant publication bias (P = 0.630). However, considering the small number of studies, the results of Egger's test should be interpreted with caution.

4. Discussion

This meta-analysis evaluated data from nine studies (6,180,434 participants) exploring whether prenatal antibiotic exposure influences the development of ADHD in offspring. The findings confirmed an

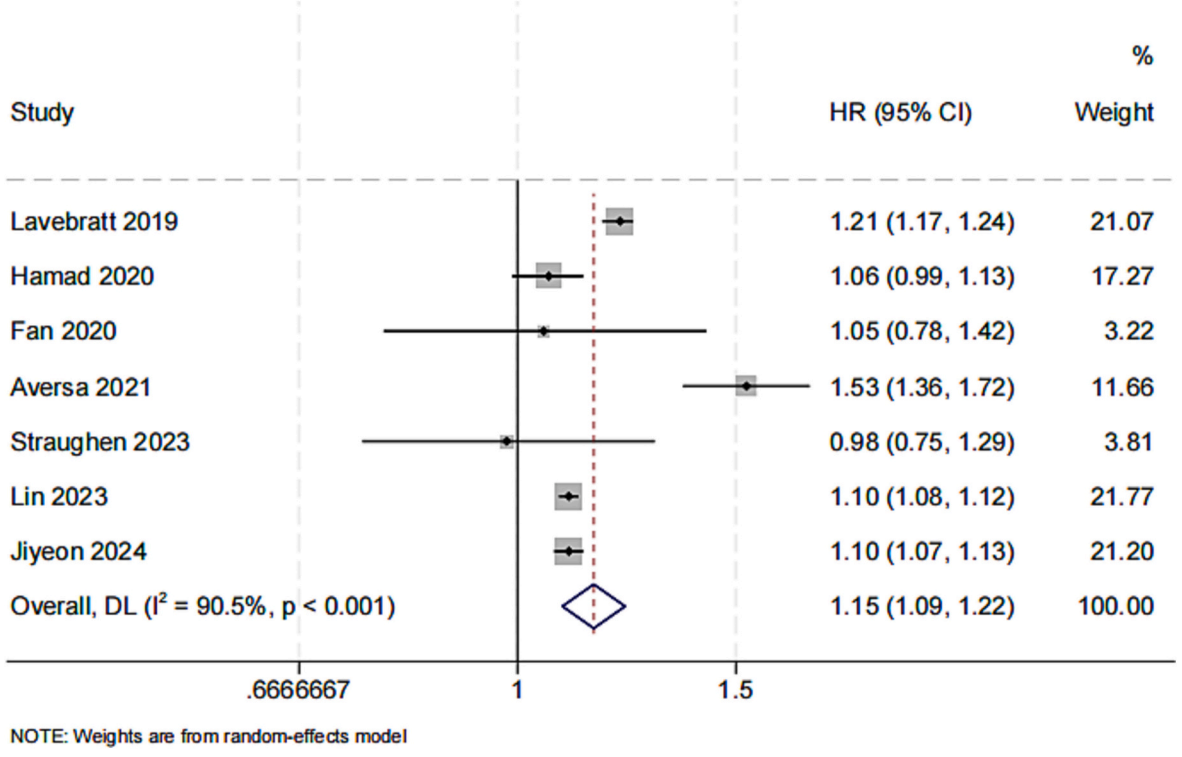


Fig. 2. Forest plot of the association between antibiotic use during pregnancy and childhood ADHD.

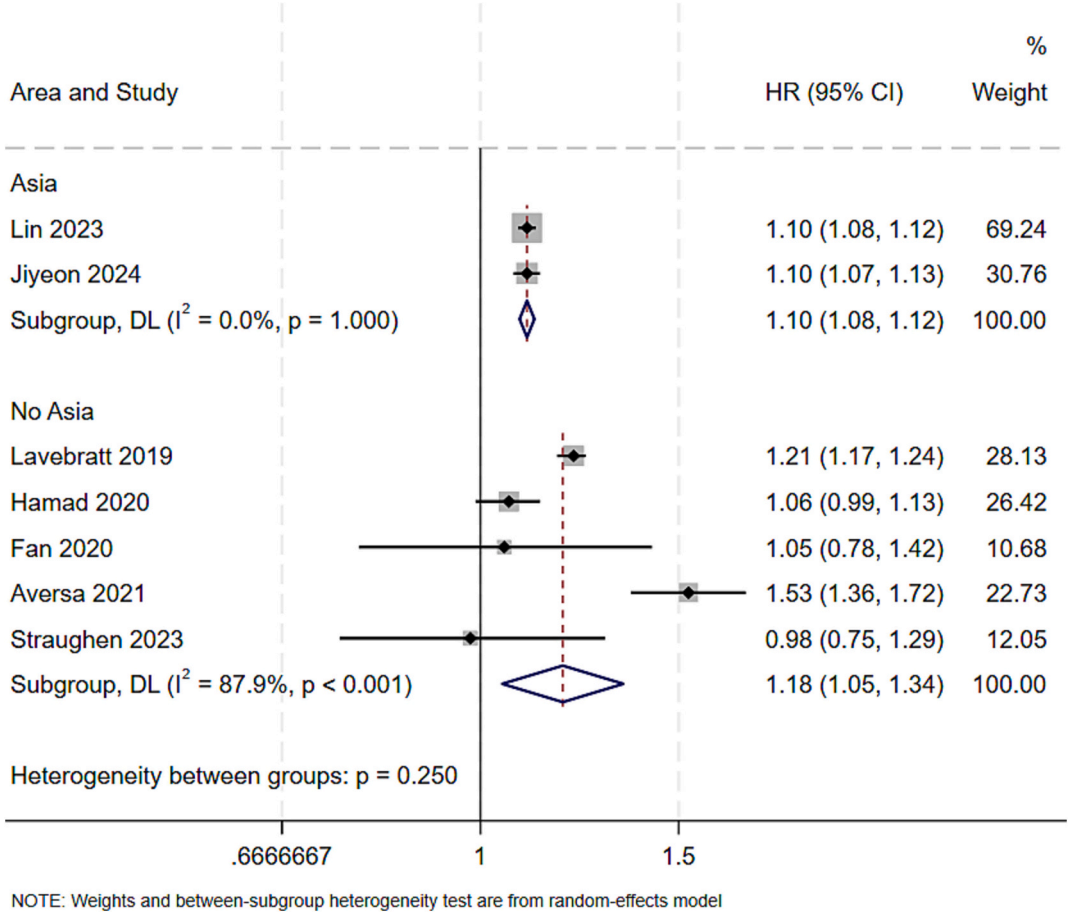


Fig. 3. Forest plot of subgroup analysis of the association between antibiotic use during pregnancy and childhood ADHD.

association between prenatal antibiotic exposure and an elevated risk of ADHD. Specifically, antibiotic exposure during either mid- or late pregnancy was associated with increased risk of ADHD. Repeated antibiotic exposures (≥2 times) also were associated with increased risk.

Previous meta-analyses (Ai et al., 2021; Tao et al., 2022; Otten et al., 2022) have investigated how prenatal antibiotic exposure (Hamad et al., 2020) affects offspring psychiatric disorders. The current analysis, incorporating recent cohort studies, corroborates these findings. Moreover, our study includes a larger sample size, adds Asian populations, focuses exclusively on ADHD without confounding psychiatric conditions, and additionally analyzes the association between (Fan et al., 2020) ADHD and exposure timing and frequency.

Ai et al. (Ai et al., 2021) reported associations between antibiotic exposure across different gestational periods and an increased risk of ADHD in offspring. The present analysis yielded slightly different results, showing no association between early pregnancy exposure and ADHD risk. Among the studies included in our analysis, five (Hamad et al., 2020; Fan et al., 2020; Straughen et al., 2023; Lin et al., 2023; Jiyeon et al., 2024) stratified antibiotic exposure by gestational period. Three studies (Hamad et al., 2020; Fan et al., 2020; Straughen et al., 2023) found no significant association between gestational antibiotic exposure and offspring ADHD risk, whereas two studies (Lin et al., 2023; Jiyeon et al., 2024) found that exposure during any pregnancy stage was associated with increased risk. The inconsistent results may be related to differences in sample size, geographical region, control of confounding factors, antibiotic classes used, and study outcomes. When pooling the relevant results, no association was observed between early pregnancy antibiotic exposure and ADHD risk, whereas mid-to-late pregnancy exposure was associated with increased risk. The study by Jiyeon et al. (Jiyeon et al., 2024) had a large sample size and greater statistical power, potentially enabling detection of subtle but significant associations and thus exerting greater influence on the pooled results. It should be noted that studies by Fan et al. and Lin et al. (Fan et al., 2020; Lin et al., 2023) used incomplete classification of antibiotic exposure periods, which may have influenced our findings. The absence of a significant association between early pregnancy antibiotic exposure and ADHD risk raises questions regarding the influence of such exposure during critical neurodevelopmental windows. Future research should consider maternal stress, nutritional status, and mode of delivery (cesarean versus vaginal birth) as potential influencing factors, and further explore antibiotic use in early pregnancy.

The mechanisms underlying the association between gestational antibiotic exposure and increased offspring ADHD risk may involve antibiotic-induced dysbiosis effects on the gut–brain axis (GBA). Antibiotic-induced dysbiosis can influence brain cognition and function through several pathways: modulating neurotransmitter synthesis (Chen et al., 2021), short-chain fatty acids (SCFAs) (Steckler et al., 2024), and tryptophan metabolism (Cavaleri et al., 2024); disrupting intestinal and blood–brain barrier integrity, leading to increased systemic inflammation affecting the central nervous system (Lewis et al., 2025); and modulating vagal nerve function, thereby influencing brain activity (Kim et al., 2020). Germ-free animal experiments have demonstrated that maternal microbiota changes during pregnancy can influence neonatal brain development (Castillo-Ruiz et al., 2023), and that germ-free animals exhibit marked alterations in brain function during development (Delgado-Ocaña and Cuesta, 2024), resulting in abnormal neurodevelopment and behavior. Similar patterns have been observed in antibiotic-treated mice compared to germ-free mice (Li et al., 2022), suggesting that dysbiosis can affect the nervous system.

Some studies have found that maternal infection during pregnancy is

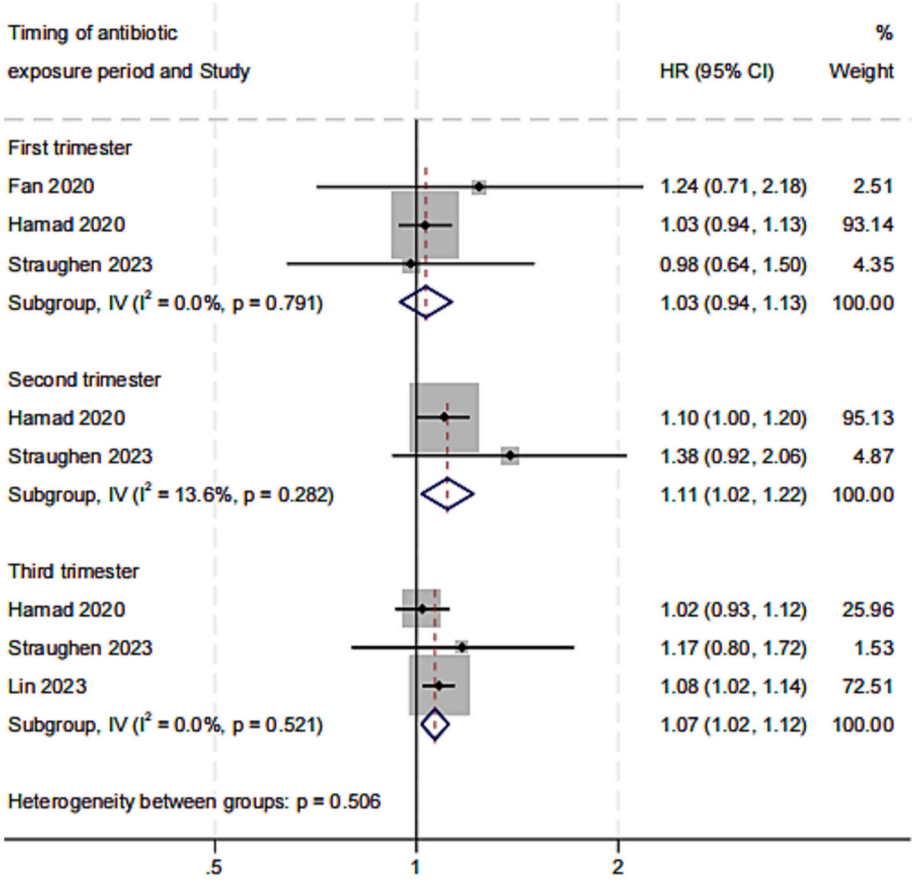


Fig. 4. Forest plot of the association between antibiotic use during different gestational periods and childhood ADHD.

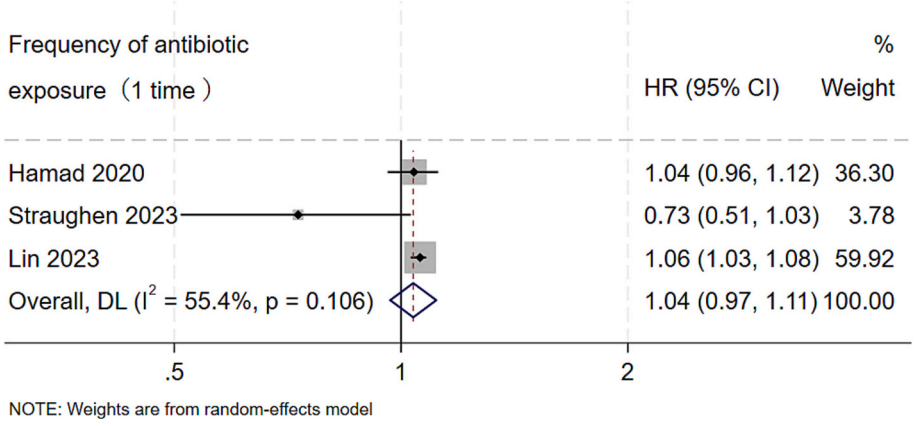


Fig. 5. Forest plot of the association between antibiotic single exposure during pregnancy and childhood ADHD.

associated with increased offspring ADHD risk (Ayubi and Mansori, 2022), potentially through maternal inflammatory responses and immune activation that affect fetal neurodevelopment. However, other studies have reported no overall association between maternal infection and offspring ADHD (Werenberg Dreier et al., 2016), with observed associations possibly attributable to inadequately controlled confounding (Zhu et al., 2022). In our review of included studies, Lavebratt et al. (Lavebratt et al., 2019) reported that adjusting for maternal infection during pregnancy did not substantially alter the estimated risk of psychiatric disorders in offspring following gestational antibiotic exposure (HR ≈ 1.16), suggesting that antibiotics may exert an independent effect. However, the outcome in that study encompassed psychiatric disorders beyond ADHD. Animal studies have shown that antibiotic exposure alone can alter offspring behavior, further supporting an independent role of antibiotics (Otten et al., 2022). Other included studies could not fully disentangle the direct effects of antibiotics from those of infection on ADHD risk. Future prospective studies should incorporate stratification by infection type, severity, and treatment modality to differentiate the independent effects of antibiotics and infection.

Jiyeon et al. demonstrated that the number of prenatal and postnatal antibiotic exposure is associated with an associated with of neuropsychiatric disorders in children (Jiyeon et al., 2024). Similarly, Z. Aversa et al. found that girls who received three to four antibiotic courses during childhood had an increased risk of ADHD compared to those with

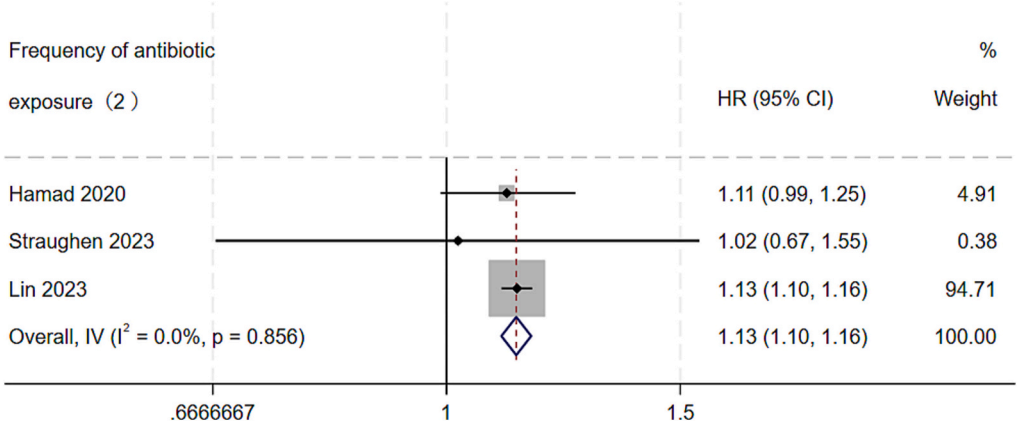


Fig. 6. Forest plot of the association between antibiotic two exposures during pregnancy and childhood ADHD.

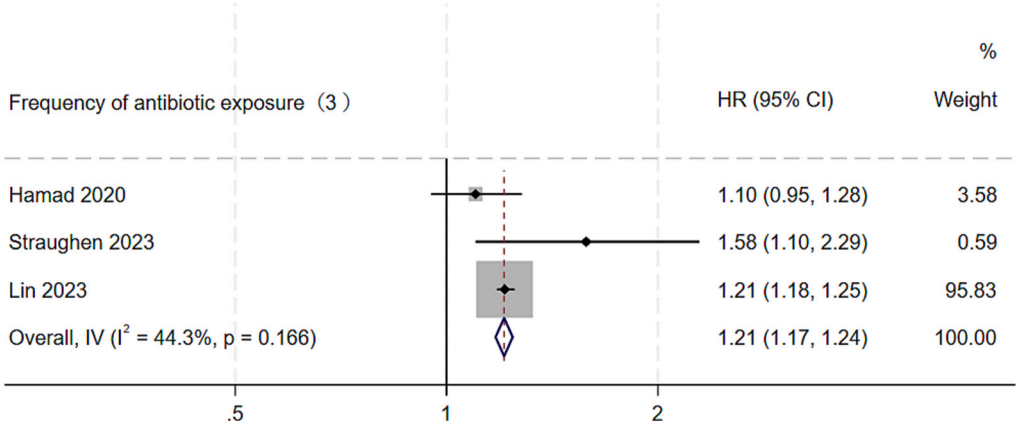


Fig. 7. Forest plot of the association between antibiotic three or more exposures during pregnancy and childhood ADHD.

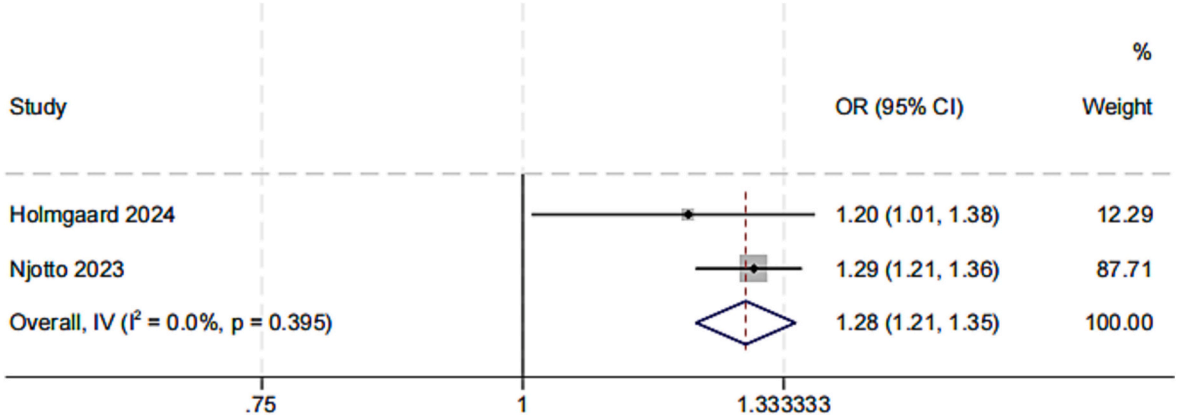


Fig. 8. Forest plot of the association between antibiotic use during pregnancy and childhood ADHD (OR).

fewer courses, while children who received five or more antibiotic courses showed significantly higher risks of hyperactivity disorder (Aversa et al., 2021). The current analysis confirms that repeated antibiotic exposures (≥2 times) are associated with an increased risk of ADHD. Multiple exposures may cumulatively disrupt the microbiota; however, the precise mechanisms require further experimental and clinical investigation. These findings suggest that clinicians should consider potential long-term neurodevelopmental effects when prescribing repeated courses of antibiotics in pregnant woman.

5. Strengths and limitations

This meta-analysis has several strengths. First, we included broader and more recent literature and extracted comprehensive data. Second, this study additionally included Asian populations. Third, stratified analyses of exposure timing and frequency were performed.

However, several limitations should be noted. Notably, due to limited primary data, it is impossible to analyze antibiotic types, exposure duration, or spectrum (broad vs. narrow) and frequency during different pregnancy periods. The absence of data on antibiotic class/type

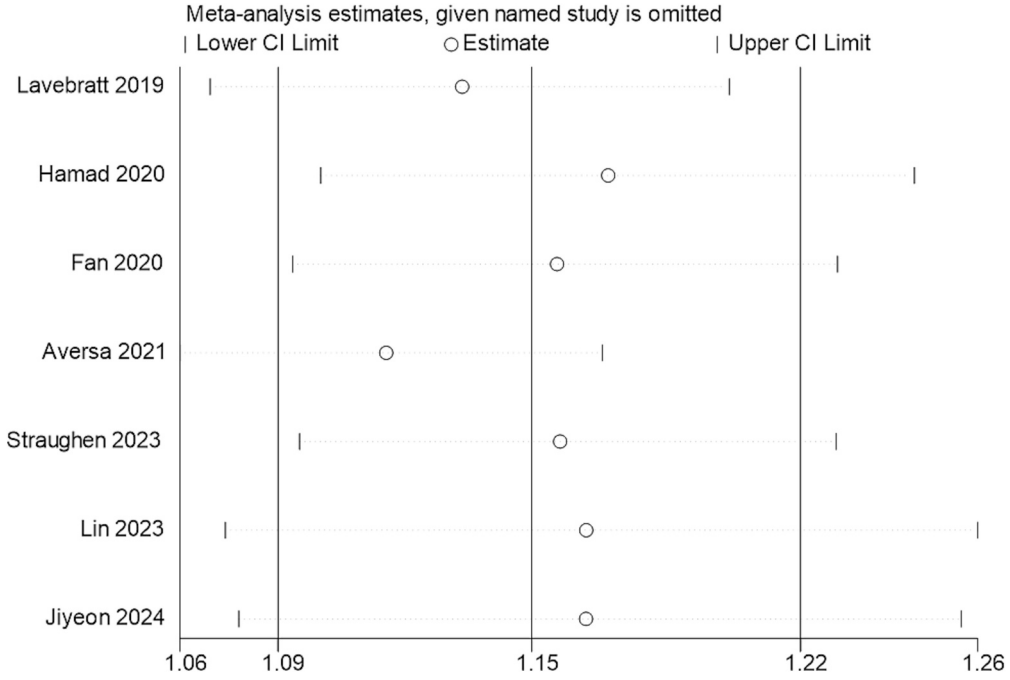


Fig. 9. Chart of stability analysis of studies on the association between antibiotic use during pregnancy and childhood ADHD.

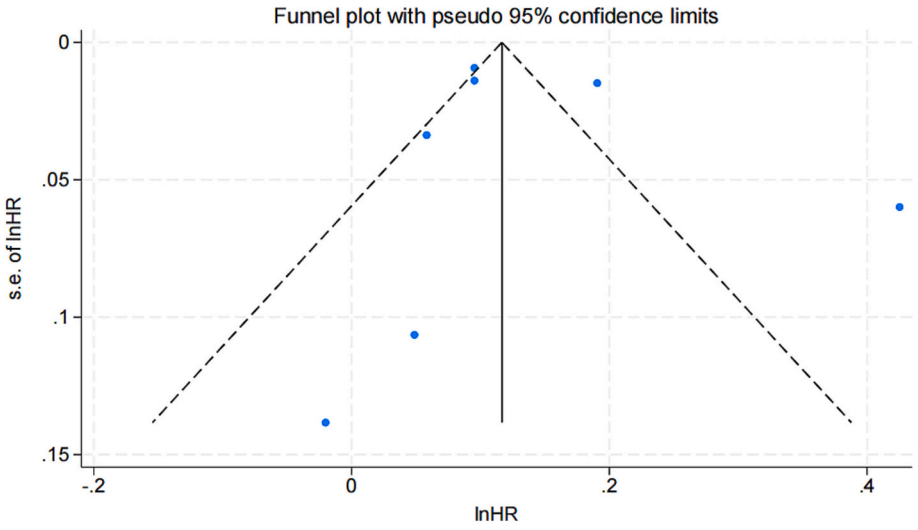


Fig. 10. Funnel plot of studies on the association between antibiotic use during pregnancy and childhood ADHD.

is a major gap. Future studies should differentiate the effects of various antibiotic classes, such as macrolides and β-lactams, on the microbiota. Second, heterogeneity was found in our results. We tried to identify the sources of heterogeneity from antibiotic types, administration timing, population characteristics, and study design. We attempted to perform meta-regression analyses (e.g., by sample size, publication year, and confounder adjustment); however, these could not be conducted due to the limited number of studies. Nonetheless, no source of heterogeneity was determined. Additionally, only English-language publications are included in our study, which may have led to selection bias. Future research should provide more comprehensive information and apply meta-regression to investigate the influence of different factors on outcomes.

6. Conclusion

This meta-analysis unveils that prenatal antibiotic exposure is associated with an increased risk of childhood ADHD, especially exposure during mid- or late pregnancy periods Repeated antibiotic exposures is associated with an increased risk. These findings indicate that antibiotic use during pregnancy should be carefully evaluated to avoid adverse effects on gut microbiota and unnecessary use.

Future research should comprehensively record prenatal antibiotic use, including indications, precise timing, classes (macrolides, beta-lactams, etc.), dosage, and duration. This comprehensive approach would facilitate the distinction between the effects of antibiotic use versus underlying conditions and enable dose-response analyses. Moreover, prospective study designs - such as sibling comparison, Mendelian randomization, or instrumental variable analysis - are warranted to explore the causal relationship between prenatal antibiotic use

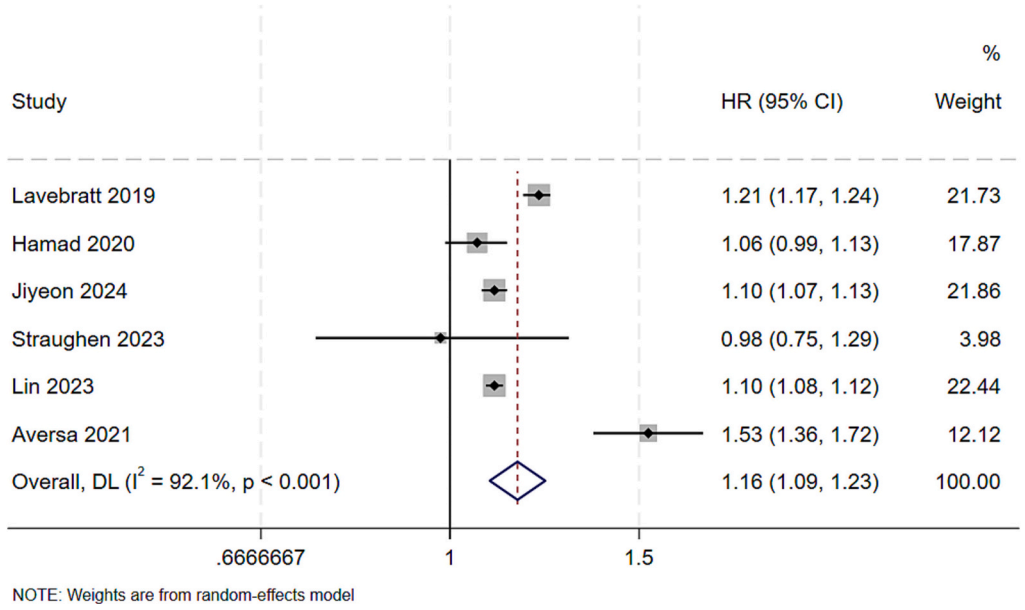


Fig. 11. Sensitivity analysis without adjusting for infection factors.

and offspring ADHD risk.

CRediT authorship contribution statement

Jiali Fan: Writing – original draft, Software, Resources. **Shanshan Wu:** Software, Methodology, Investigation. **Chengshuang Huang:** Resources, Methodology, Investigation. **Dongqiong Xiao:** Methodology, Investigation, Funding acquisition. **Fajuan Tang:** Writing – review & editing, Visualization, Supervision, Conceptualization.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Ethics approval

Not applicable.

Author statement

We confirm that this manuscript has not been published elsewhere and is not under consideration for publication by any other journal.

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Declaration of competing interest

The authors declare that they have no competing financial interests or funding to disclose.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.120168.

Data availability

No additional data are available.

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